

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

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Wavelet Analysis

PSD Varying Time-Series

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- The trigonometric basis functions used in the Fourier transform have an infinite extent and for this reason the Fourier transform may not be the best method to analyze non periodic time series data, such as the case of a localized event (e.g., a burst that decays over some time scale so that the PSD is also varying with time).
- We can (and actually do) evaluate the PSD for finite stretches of time series in order to detect its changes.
- This approach (called *spectrogram*, or *dynamical power spectra analysis*, or even *windowed Fourier transform*) suffers from degraded spectral resolution and is sensitive to the specific choice of time series segmentation length.

Windowed Fourier transform

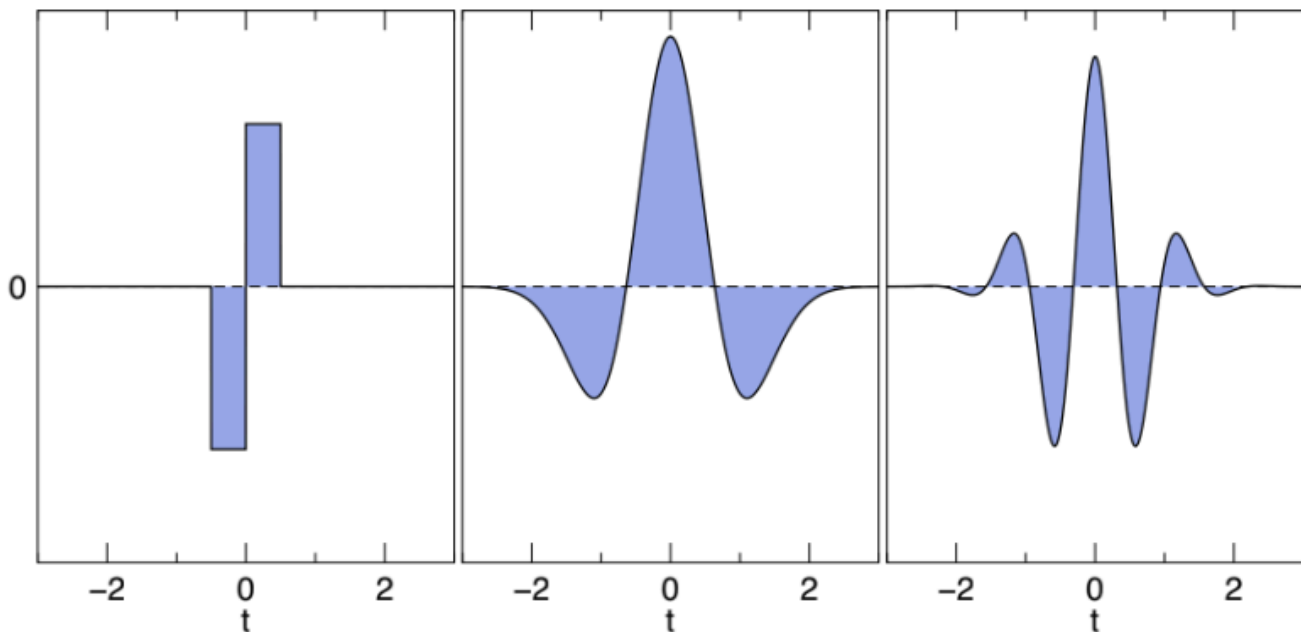
- The FT is performed on a sliding segment of length T from a time series of time step δt and total length $N\delta t$, thus returning frequencies from T^{-1} to $2\delta t^{-1}$ at each time step.
- The segments can be windowed with an arbitrary function such as a boxcar (no smoothing) or any other choice.
- The WFT represents an inaccurate and inefficient method of time–frequency localization, as it imposes a scale or “response interval” T into the analysis.
- The inaccuracy arises from the aliasing of high- and low-frequency components that do not fall within the frequency range of the window. The inefficiency comes from the $T/(2\delta t)$ frequencies, which must be analyzed at each time step.

Wavelets

- A popular family of basis functions is called wavelets.
 - By construction, wavelets are localized in both frequency and time domains.
- Individual wavelets are specified by a set of *wavelet filter coefficients*. Given a wavelet, a set of basis functions can be constructed by scalings and translations.
- Different wavelet families trade the localization of a wavelet with its smoothness. Popular wavelets include “Mexican hat”, Haar and Daubechies wavelets.
- A wavelet is a “small wave,” where small derives from the fact that it is mostly limited to an interval of time.
- A (mother) wavelet ψ has to satisfy two requirements: its integral must be zero and the integral of its square must be unity:

$$\int_{-\infty}^{\infty} \Psi(t) dt = 0 \quad \int_{-\infty}^{\infty} \Psi^2(t) dt = 1$$

- It is easy to see that these requirements imply that $\psi(t)$ is essentially non-zero only over a limited range of t and that it has to extend both above and below zero.

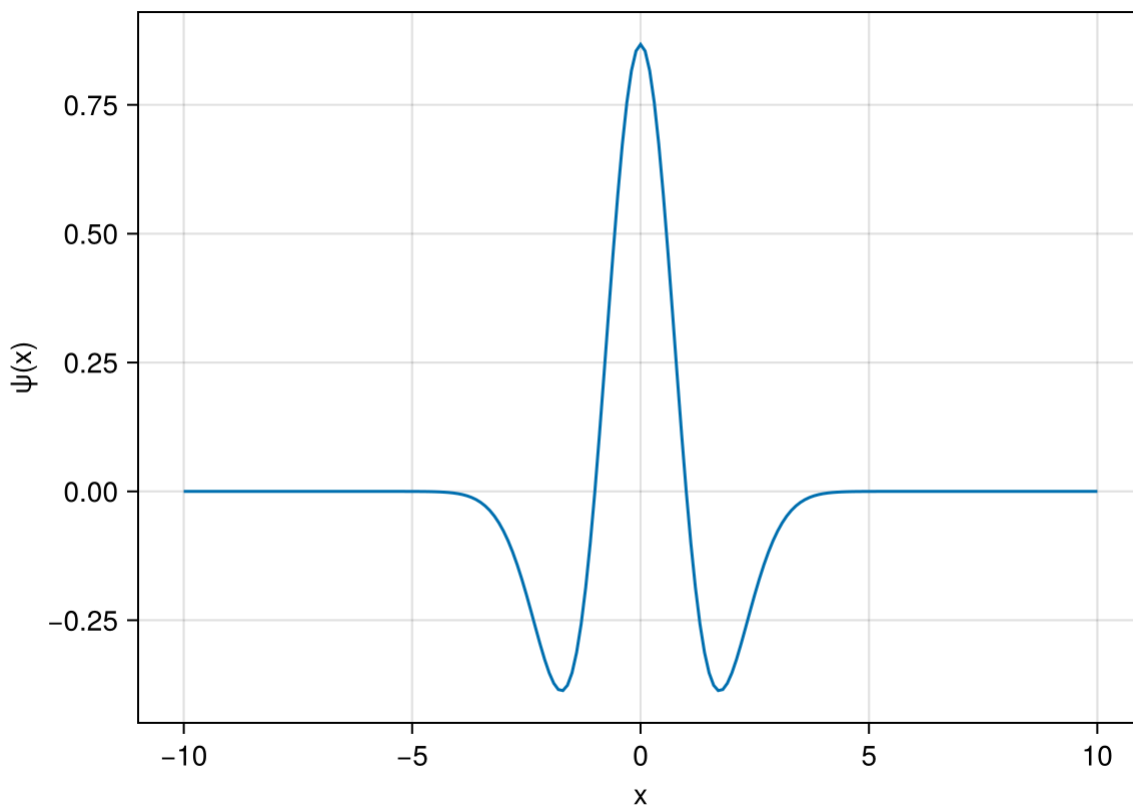


- Three examples of wavelets. From left to right: Haar wavelet, mexican hat and Morlet wavelet.

For instance, the Mexican hat function is:

$$m(x) = \frac{2}{\sqrt{3\sigma\pi^{\frac{1}{4}}}} \left(1 - \frac{x^2}{\sigma^2}\right) e^{\frac{-x^2}{2\sigma^2}}$$

Mexican Hat Wavelet



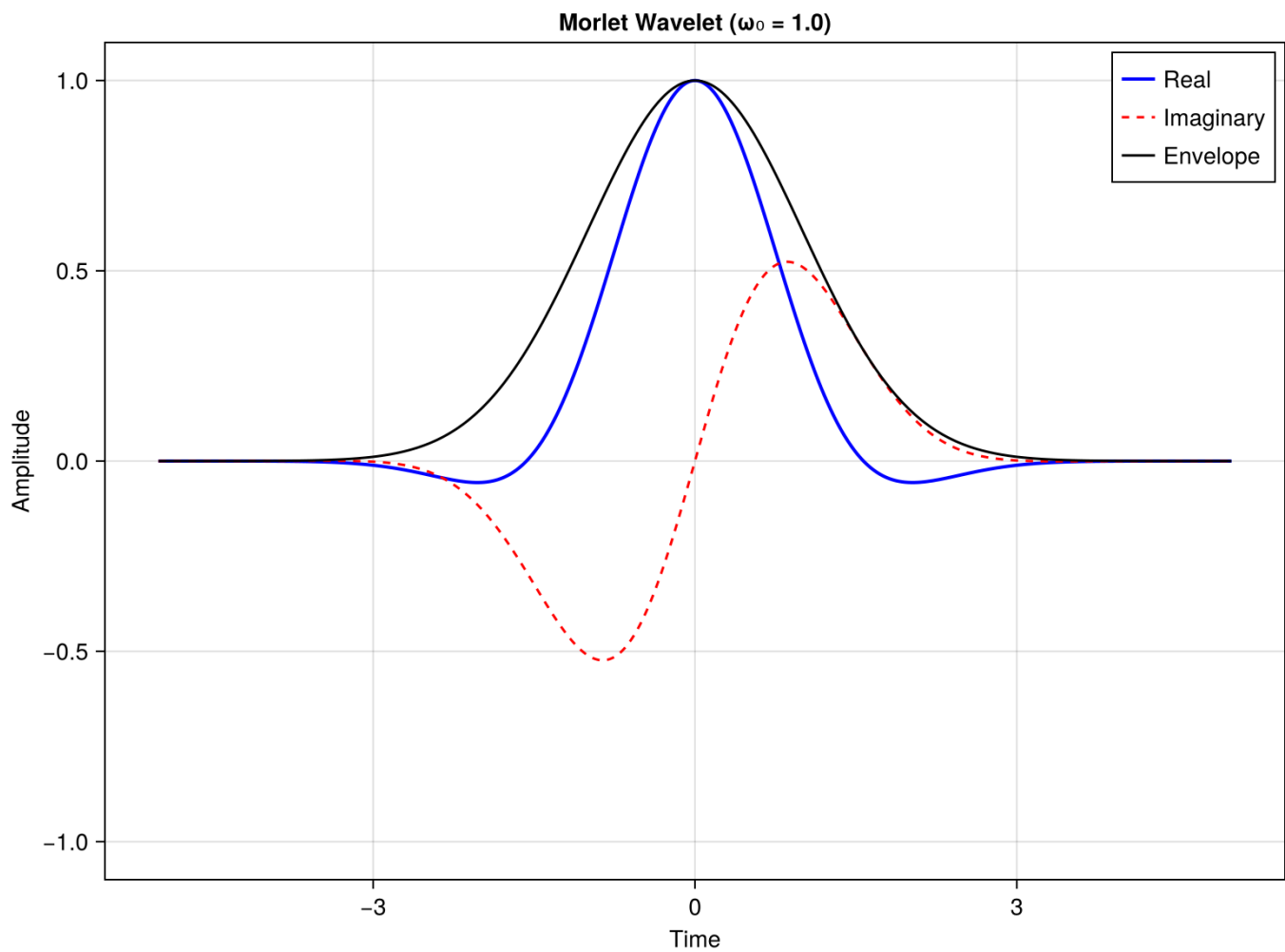
- The Mexican hat function/wavelet is actually the rescaled negative second derivative of the Gaussian function:

$$g(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}$$

- And a Morlet wavelet is nothing more than a sine wave multiplied by a Gaussian envelope:
- The mathematical expresion for the Morlet wavelet is:

$$\psi(t) = \pi^{-1/4} e^{i\omega_0 t} e^{-\frac{t^2}{2}}$$

Wavelet central frequency ω_0 :



- A continuous wavelet analysis (CWT) compares the signal to shifted and compressed or stretched versions of a wavelet. Stretching or compressing a function is collectively referred to as *dilation* or *scaling* and corresponds to the physical notion of scale. By comparing the signal to the wavelet at various scales and positions, you obtain a function of two variables.
- A wavelet can therefore be shifted by τ and dilated by a scale parameter σ :

$$\Psi_{\tau,\sigma}(t) = \frac{1}{\sqrt{\sigma}} \Psi\left(\frac{t - \tau}{\sigma}\right)$$

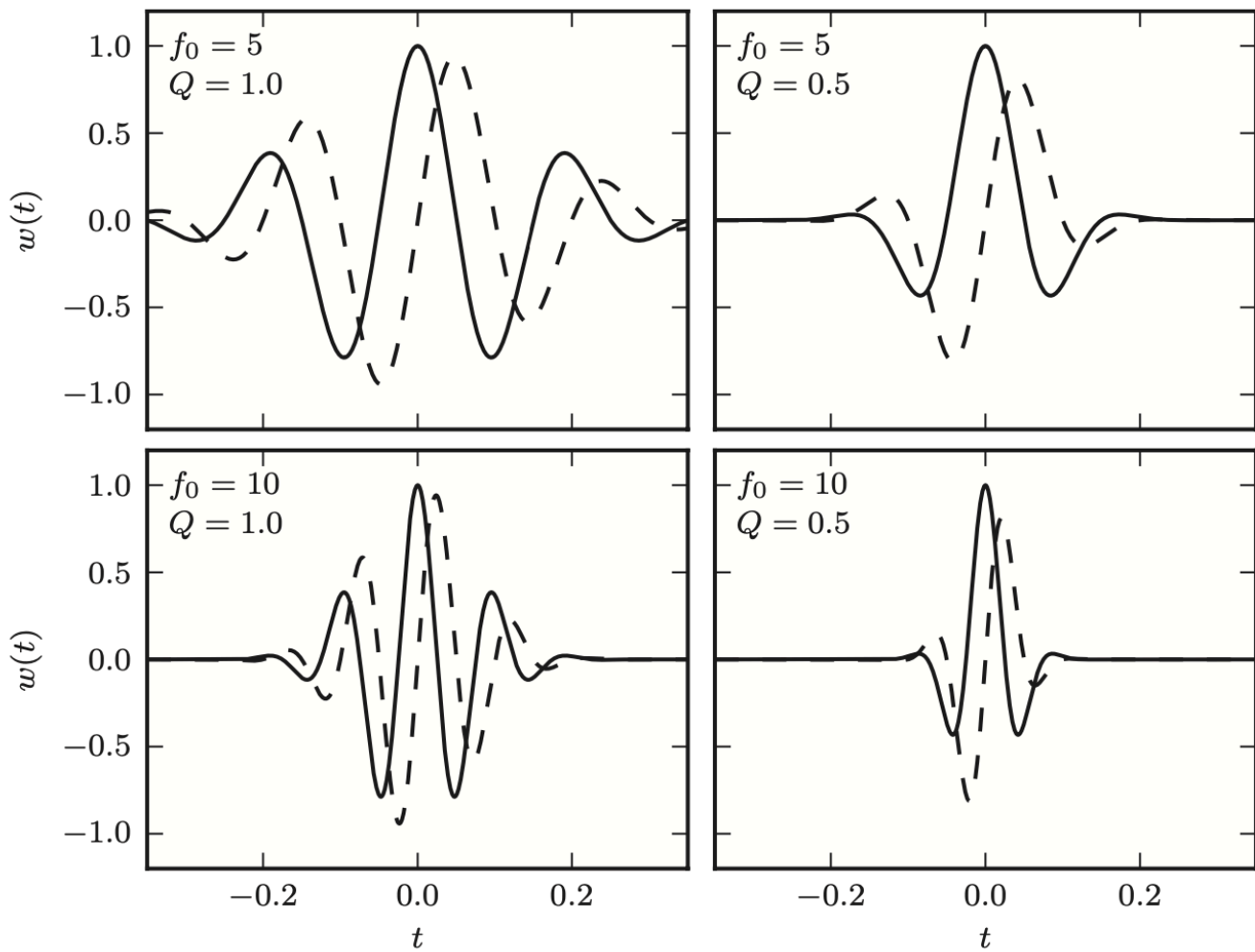
- Again $\psi(t)$ is essentially non-zero only over a limited range of t and that it has to extend both above and below zero.
- The wavelet transform of a function $f(t)$ is computed by correlating $f(t)$ with the complex conjugate of $\psi_{\tau,\sigma}(t)$ (wavelets can also be complex functions):

$$W[f(\tau, \sigma)] = \int_{-\infty}^{\infty} f(t) \Psi_{\tau,\sigma}^*(t) dt$$

- By continuously varying the values of the scale parameter, σ , and the position parameter, τ , you obtain the CWT coefficients that are typically shown by a 2D plot.

- Again referring to the Morlet wavelet, we can write it as:

$w(t|t_0, f_0, Q) = A e^{i2\pi f_0(t-t_0)} e^{-f_0^2(t-t_0)^2/Q^2}$, where t_0 is the central time, f_0 is the central frequency, and the dimensionless parameter Q is a model parameter which controls the width of the frequency window.



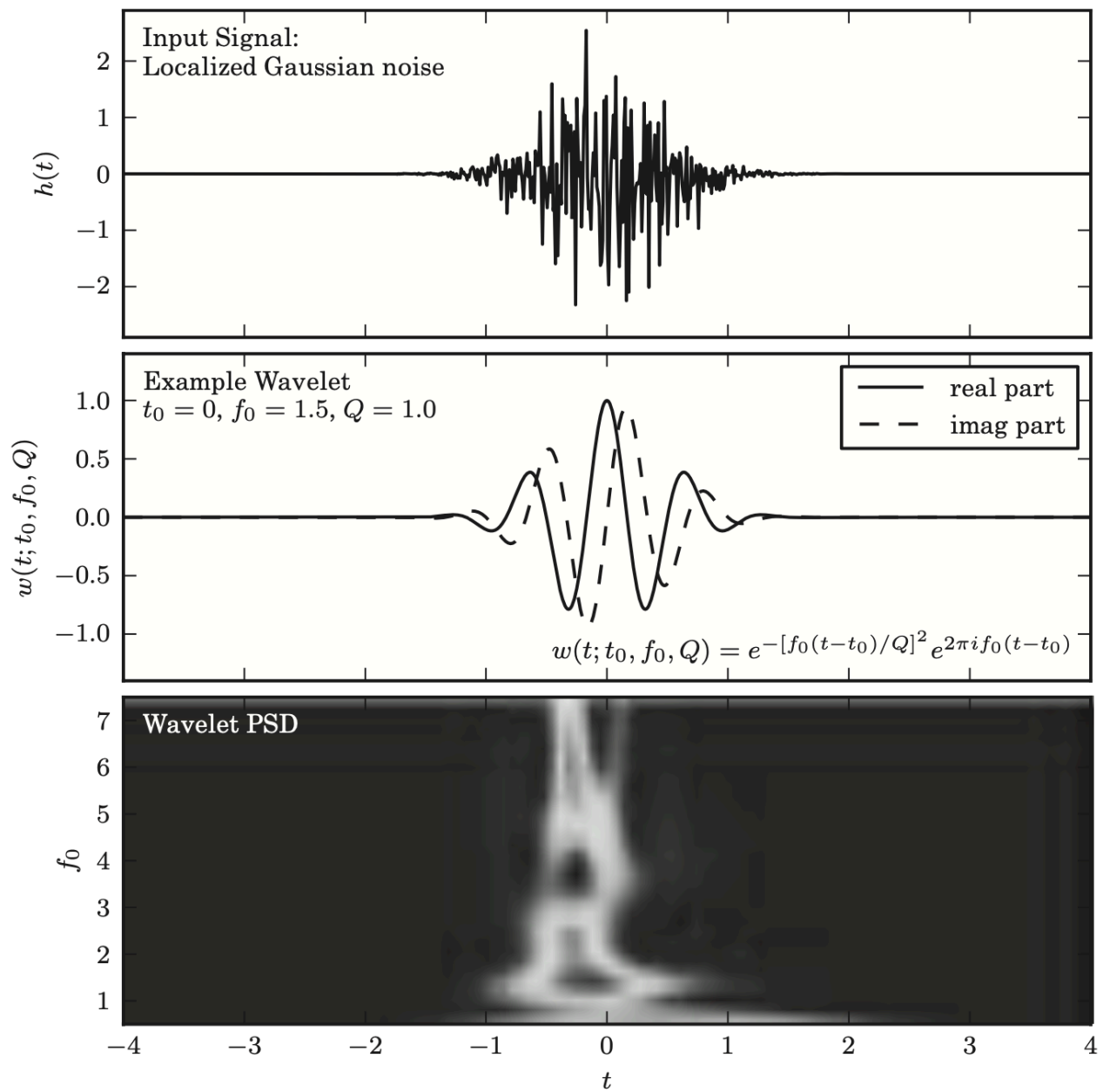
- Wavelet analysis has, roughly speaking, some connections with a “matched filters” analysis.
- Given a signal $h(t)$, its wavelet transform is given by the convolution:

$$H_w(t_0; f_0, Q) = \int_{-\infty}^{\infty} h(t)w(t|t_0, f_0, Q)$$

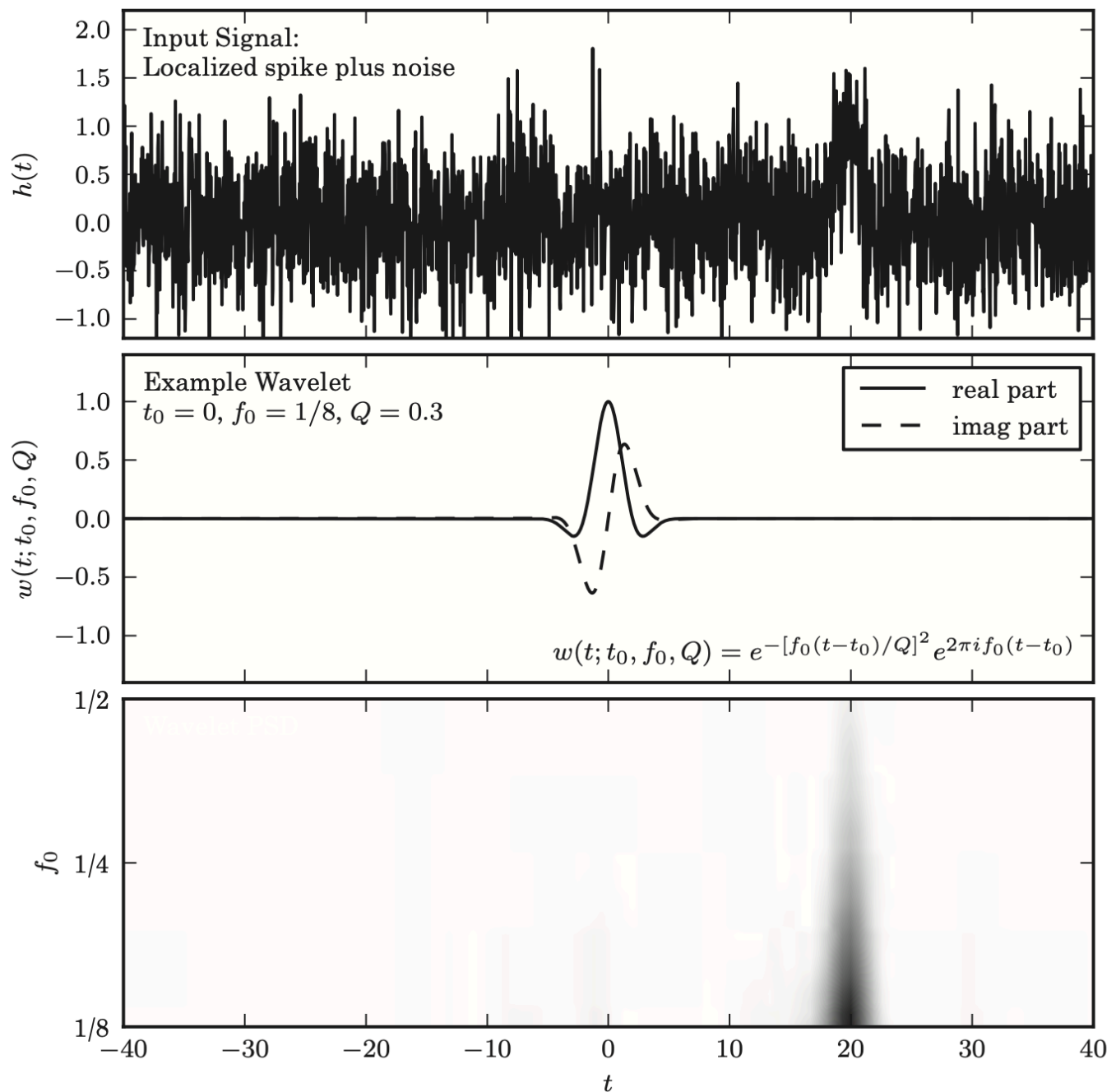
- By the convolution theorem, we can write the Fourier transform of H_w as the pointwise product of the Fourier transforms of $h(t)$ (e.g. by a DFT) and of $w(t|t_0, f_0, Q)$.
- Then, the wavelet PSD, can be defined by:

$$\text{PSD}_w(f_0, t_0; Q) = |H_w(t_0; f_0, Q)|^2$$

- As said, unlike the typical Fourier-transform PSD, the wavelet PSD allows detection of frequency information which is localized in time.
- A scaled wavelet is evaluated for various values of the scale σ (usually taken to be multiples of the lowest possible frequency), as well as all values of τ between the start and end dates. A two-dimensional picture of the variability can then be constructed by plotting the wavelet amplitude and phase.
- This is indeed one the approaches (we are simplifying a lot here...) used in many fields looking for transients events (climatology, gravitational wave astronomy, etc.).
- Example with a localized Gaussian noise:



- The upper panel shows the signal. The middle panel shows an example wavelet and the lower panel shows the power spectral density as a function of the frequency f_0 and the time t_0 , for $Q = 1.0$.
- And now a different example:



- The upper panel shows the input signal, which consists of a Gaussian spike in the presence of white (Gaussian) noise. The middle panel shows again an example wavelet. The lower panel shows the PSD as a function of the frequency f_0 and the time t_0 , for $Q = 0.3$.

Let's see a wavelet "in action":

- We show step by step how a *Morlet wavelet* slides along a signal and produces the *convolution coefficients* (continuous wavelet transform).

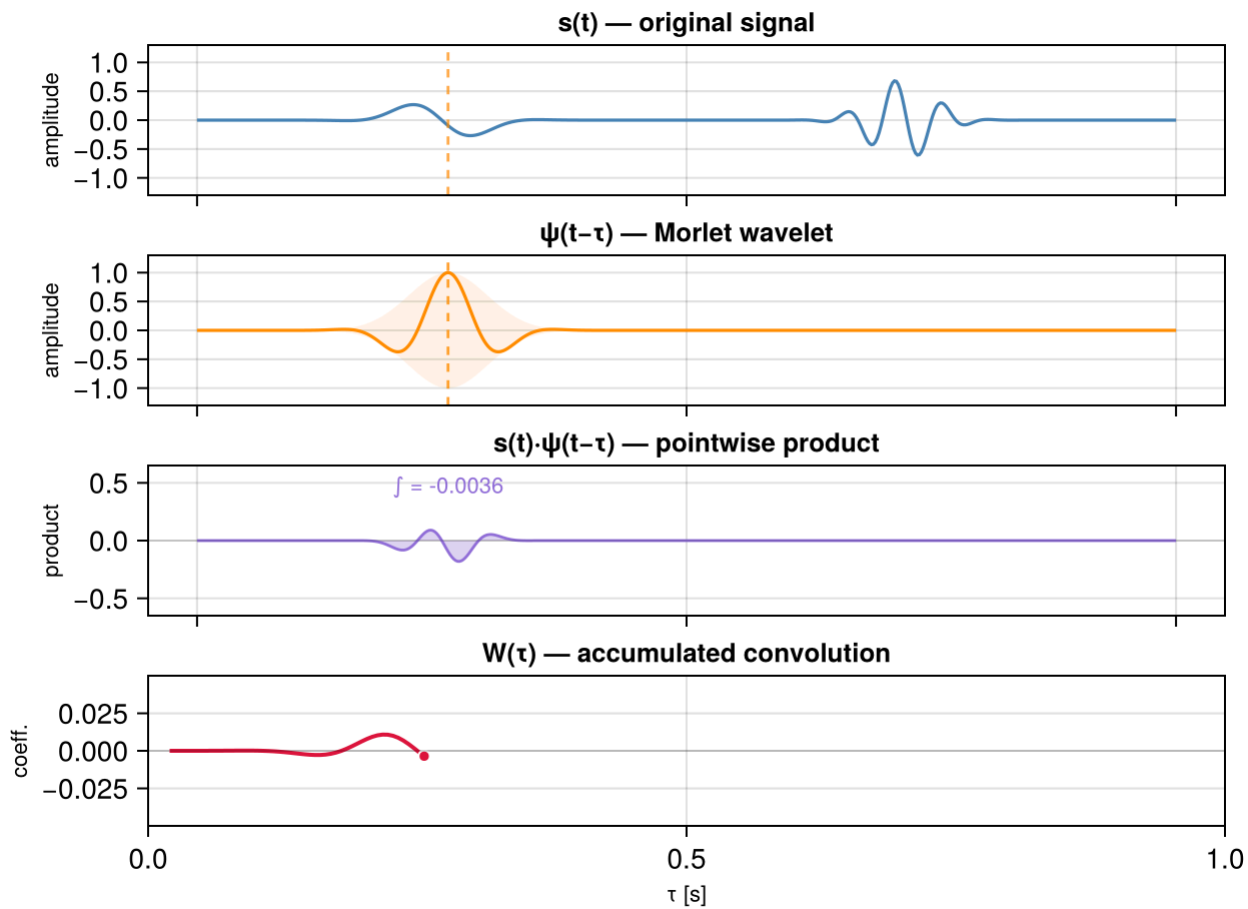
Wavelet position τ (s):



Central frequency f_0 (Hz):



Width σ of the Gaussian envelope (time window):



- When the wavelet overlaps a burst at the same frequency as f_0 , the pointwise product is consistently positive.
- Increasing f_0 from 8 to 20 Hz we see the first burst (6 Hz) disappearing, while the second emerges.
- Decreasing σ we have a narrower window that gives better time resolution but poorer frequency resolution.

Exercise: wavelet analysis of a chirp signal

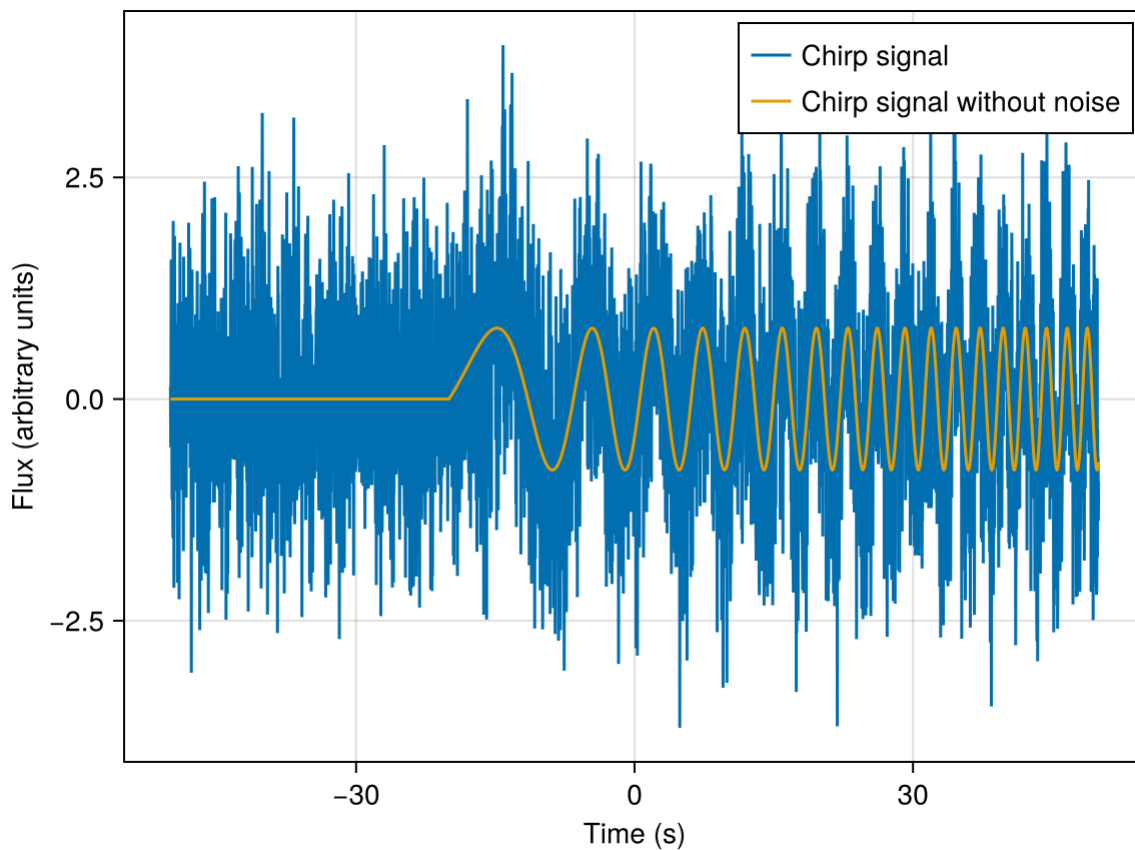
- Let's define a "chirp" signal:

```

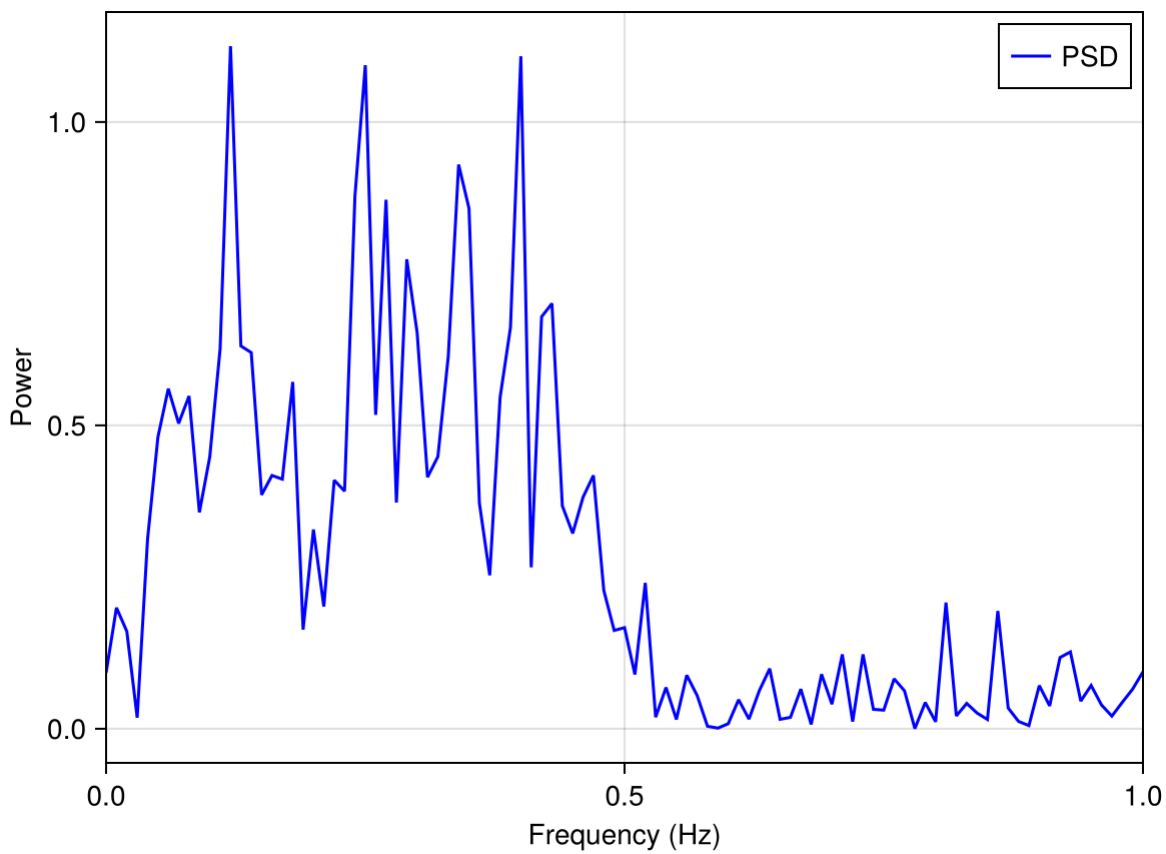
1 begin
2   function chirp(t, T, A, phi, omega, beta)
3     signal = A .* sin.(phi .+ omega .* (t .- T) .+ beta .* (t .- T).^2)
4     signal[t .< T] .= 0
5     return signal
6   end
7
8   N = 4096
9   t = range(-50,50,N)
10  h_true = chirp(t, -20, 0.8, 0, 0.2, 0.02)
11  d = Normal(0.,1.)
12  h = h_true .+ rand(d,N)
13 end;

```

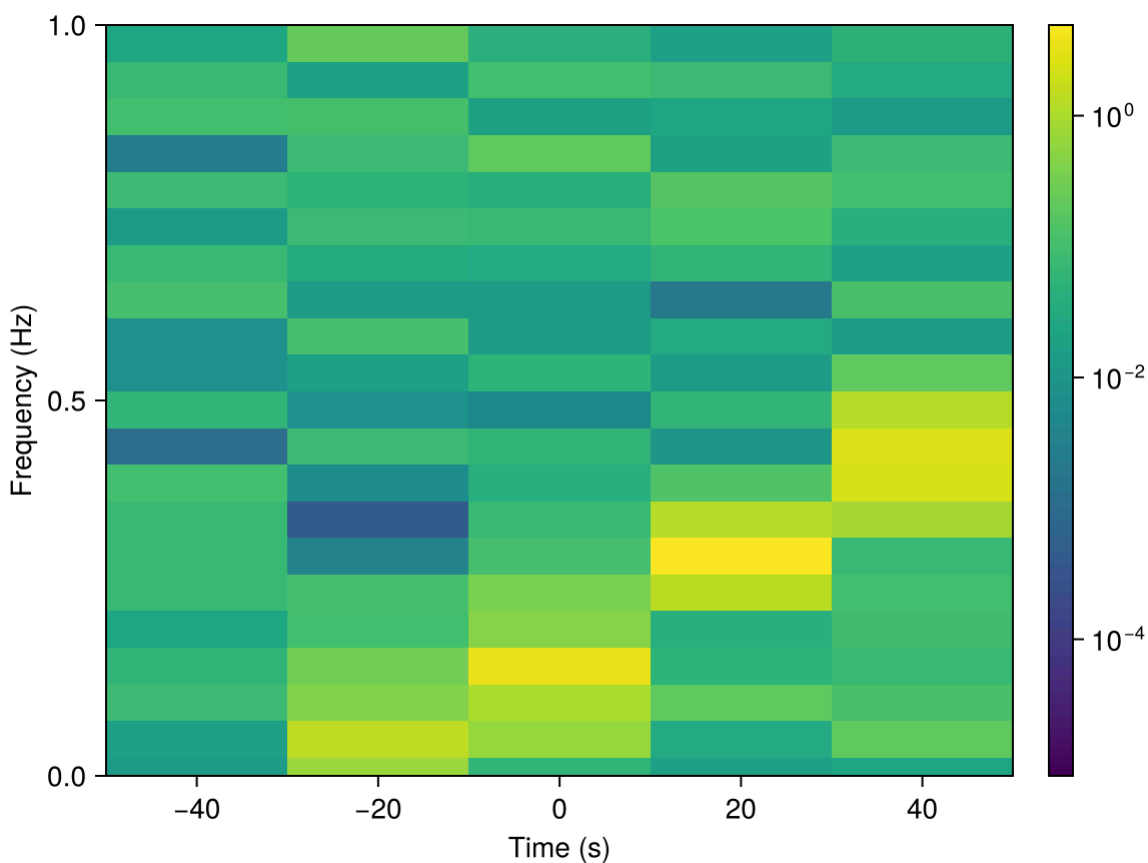
- Do you want to remove the clean signal from the plot? ☐



- Let's first compute a Fourier periodogram of this signal:

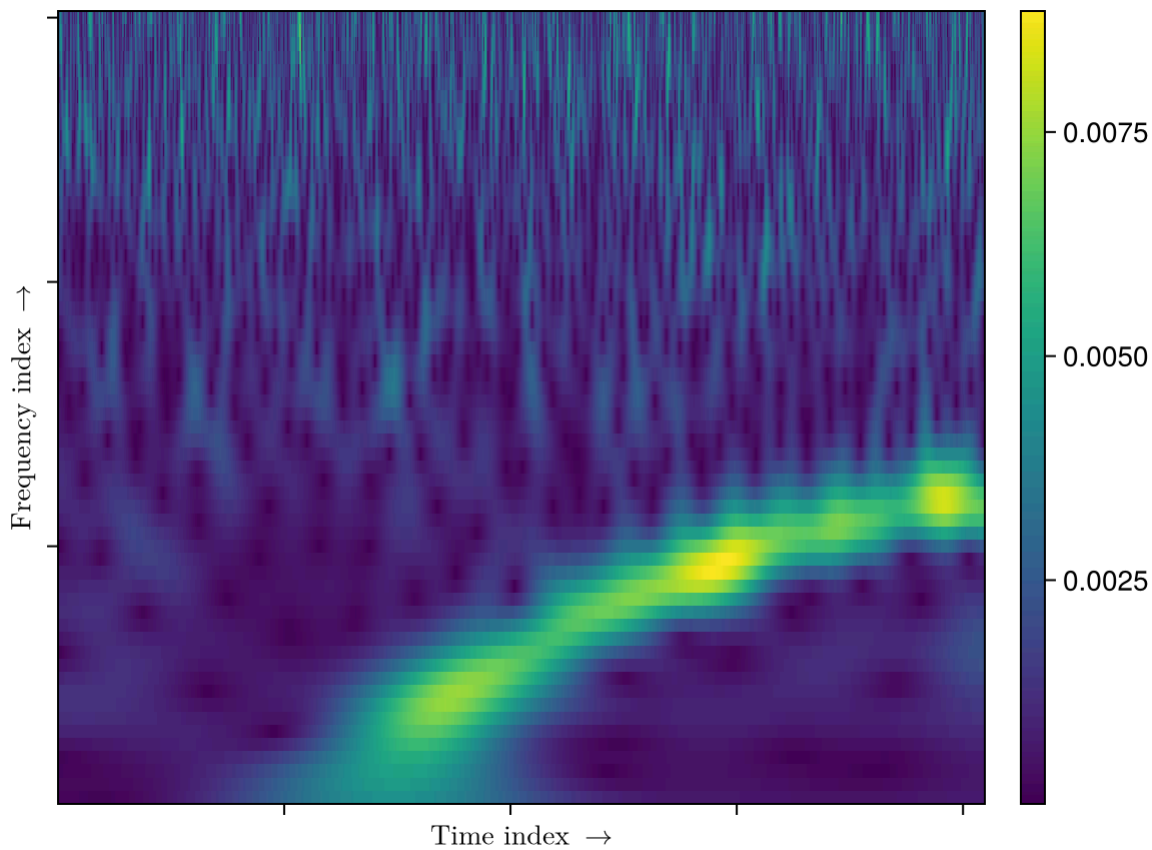


- The periodogram shows power at the lowest frequencies with a break at frequencies higher than $\sim 0.5\text{Hz}$, i.e., for time-scales shorter than $\sim 1\text{s}$.
- Let's try to study the time-evolution of the PSD by a *spectrogram*:



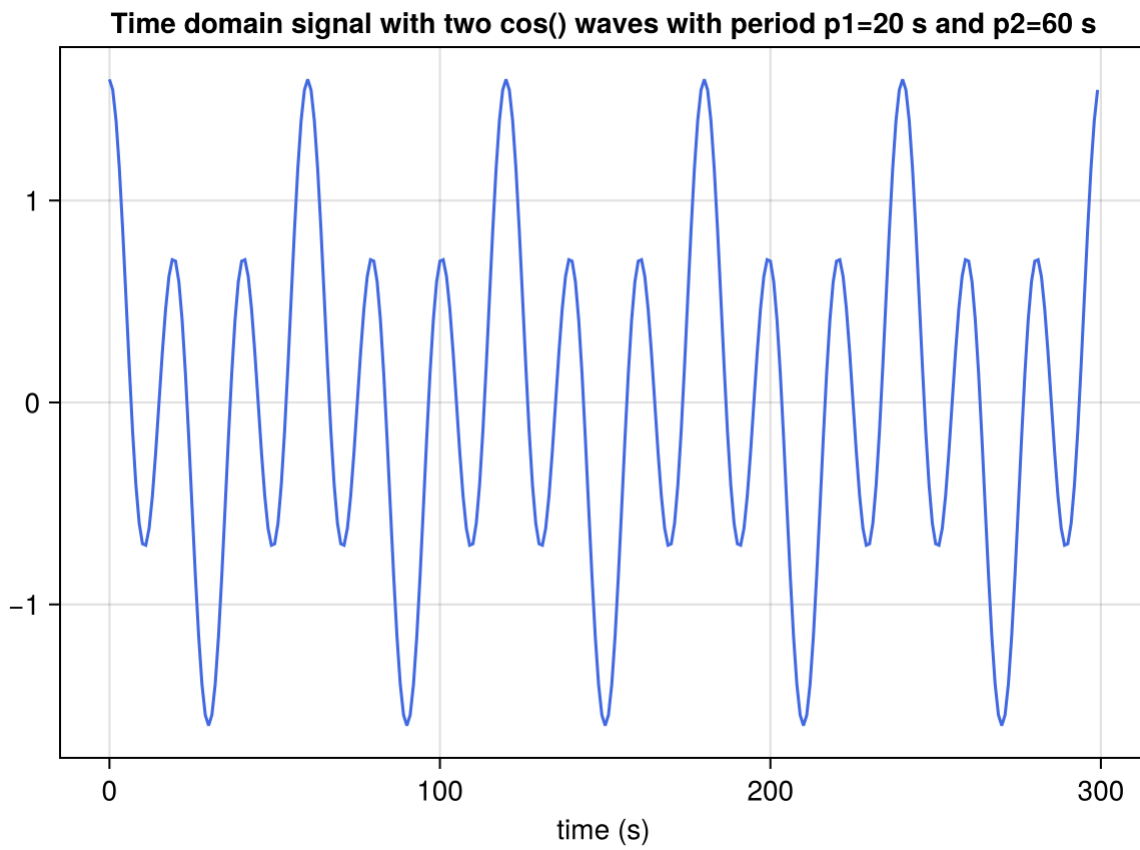
- And, it is indeed clearly visible a signal appearing at about -20s with frequency increasing with time.

- Nevertheless, this is not an optimal solution since the number of bins is arbitrary, and clearly we reduce the resolution of the analysis decreasing the length of the time-series.
- This is where a wavelet analysis can give a contribution.
- The result of a wavelet analysis is shown below:

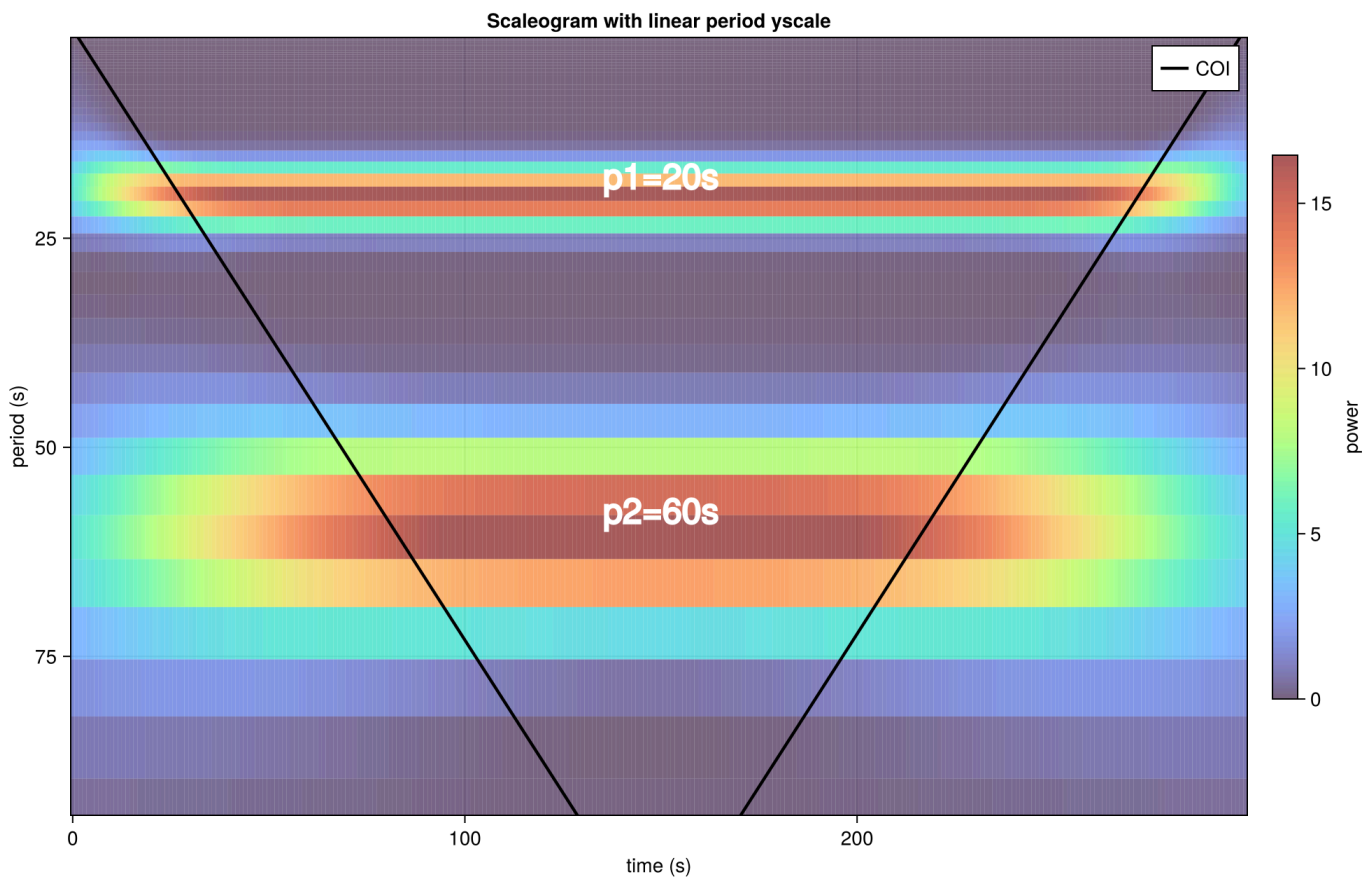


- And we can appreciate how the formation and evolution of the signal is clearly visible in the plot.

Example: data with two periodic signals



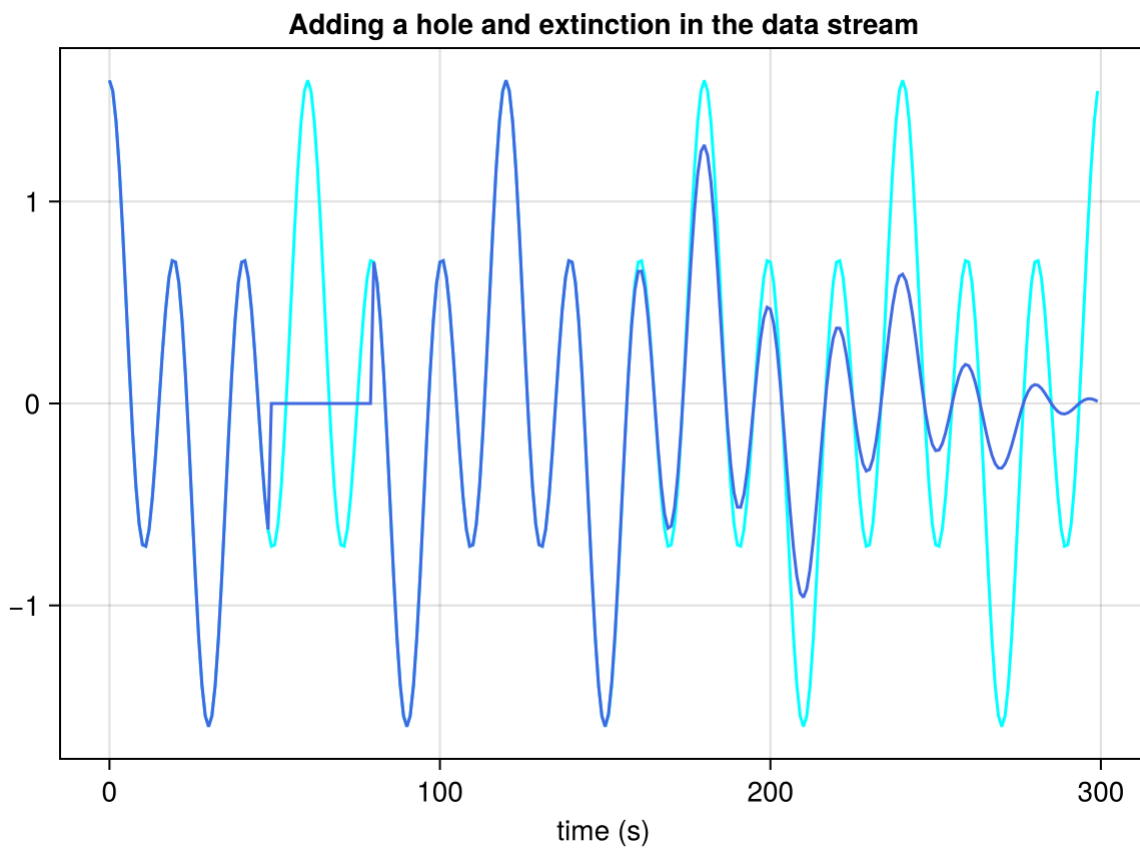
- The plot clearly shows the two signals, and that they are present for the whole observation.



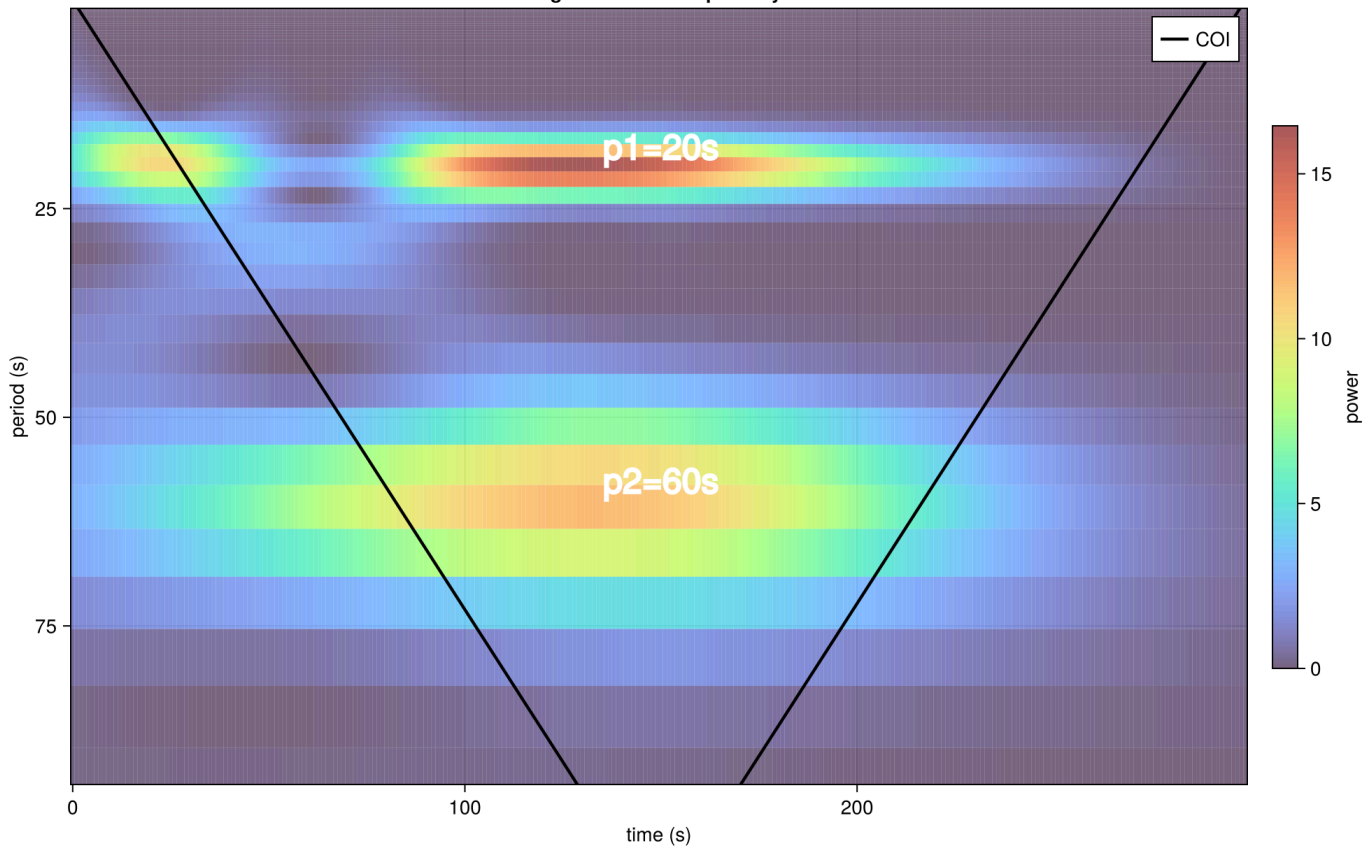
- The two periodic signals do appear as continuous stripes along the time axis (X). The wavelet captures the signal at its frequency of sensitivity.
- The visible height of the two signals expands with the scale value. This happens because the wavelet signal is scaled by a factor s .
- The solid black lines identify the Cone Of Influence (COI), they represent the locations where the data is strongly affected by border effects due to the extent in time of the child wavelet.

Effect of holes in data

- In real life, data has holes, the observed phenomenon amplitude can vary over time, etc.
- The following example has a hole at the beginning and a ramp to zero is applied at the end to get some hint on how these effects do appear on the *scaleogram*.



Scaleogram with linear period yscale

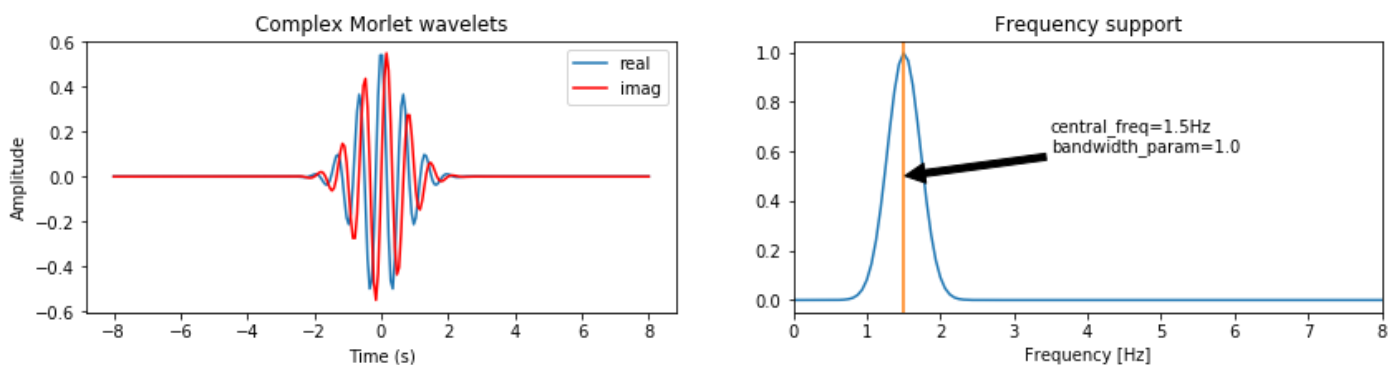


- As you can see, the scale influences not only the sensitivity on the Y axis but has also a smoothing effect on the X axis: the second `cos()` line at period does not "see" the first hole in the data, but the signal amplitude has diminished.
- The second observation is the appearance on a weak spurious line at period ~30. This is due to the sharp cut applied on the data between 50 and 100. This effect is another example of aliasing in frequency.

Wavelets: a few more technical concepts

What is the *scale* parameter?

- There could be some confusion between the *scale* of a wavelet and the *frequency*, the *scale* matches the extent over time of the wavelet.
- Essentially, a mother wavelet function is built and has an extent over time of a given *base duration*, bs and the Continuous Wavelet Transform (CWT) is performed for each value of *scales*, s , with a child wavelet of duration $bs \times s$.
- This child wavelet is convolved with the data array.
- The mother wavelet is described by the function of time $\Psi(t)$
- The child function at scale s is $\Psi_{s,x}(t) = \frac{1}{\sqrt{s}} \frac{\Psi(t-x)}{s}$, where x is the shift parameter used to convolve the signal being analysed with the wavelet.
- The frequency, ω , associated to the scale value s is $\omega = \frac{\omega_0}{s}$, where ω_0 is the parameter of the chosen wavelet.
- The central frequency corresponds to the signal frequency to which the wavelet function will be the most sensitive for scale $s = 1$.



- In the right plot (Frequency), the yellow vertical line shows the location of the peak: this is the *central frequency* of the wavelet.
- The bandwidth parameter selects how much the wavelet is sensitive to the frequencies around the central one. On the frequency plot, the bandwidth is represented by the width of the bell shaped curve (also called the *wings*).

So, which period value correspond to a given *scale*?

- Wavelet functions are designed to be compact in time and frequency. When looking at the plot above, you can see that the oscillations fall quickly around the center of the function. This *compactness in time* allows the scalogram to show local measurements of the signal *variations*. In fact, from the considerations above: ***period*** $= \frac{s}{\omega_0}$
- Of course, please pay attention that, at a given scale, the wavelet sensitivity is maximum to the central frequency but the bandwidth is not null, hence it will also probe frequency around the *central frequency* value.
- This will appear on scalogram as a smoothing effect along the Y axis. This property will also manifest in time domain where the signal decreases *smoothly* to zero around the maximum oscillation amplitude.

For this reason it is probably better to use the term of *pseudo period* to describe the time scale. This property manifests on the *scaleograms* with some amount of fuzziness around the signal

features which are well localized in time. This fuzziness is proportional to the scale s .

- Another important factor is the choice of a set of scaling parameters s , such that we adequately sample all the frequencies present in our time series.
- We typically first choose the smallest resolvable scale, s_0 , as some multiple of our time resolution of the input data, δt .
- The larger scales (longer periods) are (often) chosen as power-of-two multiples of this smallest scale.
- A common choice for the scale could then be: $s_j = s_0 2^{j\delta_j}$, for $j = 1, \dots, J$.
- And therefore the maximum scale becomes: $J = \delta_j^{-1} \log_2(N\delta t/s_0)$.

The difference between period and scale is essentially that the scale refers to the width of the wavelet. As the scale increases and the wavelet gets wider, it includes more of the time series, and the finer details get smeared out. The scale can be defined as the distance between oscillations in the wavelet (e.g. for the Morlet), or it can be some average width of the entire wavelet (e.g. for the Marr or Mexican hat).

- The period (or inverse frequency) is the approximate Fourier period that corresponds to the oscillations within the wavelet. For all wavelets, there is a one-to-one relationship between the scale and period. The relationship can be derived by finding the wavelet transform of a pure cosine wave with a known Fourier period, and then computing the scale at which the wavelet power spectrum reaches its maximum.

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Belloni & Bhattacharya \(2022\) - "Basics of Fourier Analysis for High-Energy Astronomy"](#)
- [Ivezić et al. \(2020\) - "Statistics, Data Mining, and Machine Learning in Astronomy"](#)

Further Material

Papers for examining more closely some of the discussed topics.

- [Torrence & Compo \(1998\) - "A Practical Guide to Wavelet Analysis"](#)

Credits

This notebook contains material obtained from
https://www.astroml.org/bookfigures/chapter10/figchirp2_PSD.html,
<https://github.com/alsauve/scaleogram/blob/master/doc/scale-to-frequency.ipynb>,
<https://paos.colorado.edu/research/wavelets/wavelet1.html>,
<https://paos.colorado.edu/research/wavelets/faq.html>.

Course Flow

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html	Science case about variable stars	Science case about climate data	Course Summary

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